
Speech Tokenizers are Vulnerable: Transferable Semantic Attack and Robust Tokenizer

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Abstract

1 Speech tokenizers serve as the critical interface between continuous speech signals
2 and discrete representations, and are widely used in speech systems, e.g., Automatic
3 Speech Recognition (ASR) and Large Audio-Language Models (LALMs). In this
4 paper, we find that speech tokenizers are fragile at the semantic level: slight
5 perturbations can disrupt their semantic encoding process and consequently cause
6 severe degradation in downstream ASR and LALM performance. To study this
7 vulnerability, we propose **T-SemAttack**, a transferable semantic attack based on
8 a semantic encoder ensemble. By jointly perturbing the representation spaces
9 of multiple semantic encoders, T-SemAttack effectively disrupts the semantic
10 content of speech while preserving perceptual quality, thereby inducing error tokens.
11 We further analyze token fragility and layer-wise representation drift in relation
12 to cross-model transferability and disruption of LALM attention. Our analysis
13 reveals a progressive amplification chain in which small waveform perturbations
14 are magnified through the tokenizer pipeline and lead to semantic collapse, e.g.,
15 the attack causes a 99.7% token change in the S3 tokenizer. Building on these
16 findings, we introduce **ROSETok**, a robust speech tokenizer that combines noisy
17 training with robust semantic distillation to improve reconstruction fidelity and
18 downstream-task robustness. Extensive experiments on multiple datasets, 9 open-
19 source and black-box ASR systems, and 4 LALMs demonstrate that T-SemAttack
20 achieves strong transferable attack performance, while the proposed Robust Speech
21 Tokenizer exhibits robustness under high-fidelity reconstruction and downstream
22 tasks. Our code and demo are available at <https://t-semattack.github.io>.

23 1 Introduction

24 In recent years, LALMs [1, 2] have demonstrated exceptional interaction capabilities and are widely
25 deployed in scenarios such as in-vehicle assistants and meeting transcription. Compared to previous
26 speech models [3, 4], current large language model (LLM)-based architectures [1, 5, 6] typically rely
27 on a speech tokenizer to map continuous signals into discrete tokens, which are subsequently fed
28 into LLMs for modeling. Consequently, the speech tokenizer bridges the continuous and discrete
29 representations, and its robustness directly dictates the reliability of the entire semantic pipeline.

30 Existing research [7, 8, 9] has primarily focused on the holistic robustness of models, while paying
31 relatively little attention to the safety issues of the speech tokenizer as the signal input module. We
32 find that speech tokenizers suffer from critical security vulnerabilities. Even slight perturbations can
33 drastically alter the encoding output, thereby affecting the LLM component’s understanding and
34 further causing severe semantic comprehension errors in subsequent ASR or LALM processing.

35 In this paper, we propose a **Transferable Semantic Attack (T-SemAttack)**, which focuses on the
36 encoder in the speech tokenizer. By jointly perturbing multiple surrogate semantic encoders [9], the

Table 1: Comparison with representative transferable attacks against ASR systems.

Method	Knowledge	Transferability	Attack Target	Bilingual	Open Source
Carlini et al. [11]	□	✗	•	✗	✓
ZQ-Attack [9]	■	✓	•	✗	✗
AdvDDoS [12]	■	✓	•	✗	✗
E2E-VGuard [7]	■	✓	•	✓	✓
AudioShield [8]	■	✓	•	✗	✓
T-SemAttack	■	✓	○	✓	✓

Legend: Knowledge: ■ Black-box, □ White-box. Attack Target: • Whole system, ○ Speech tokenizer module.

method directly corrupts the semantic content of the input speech. We employ the psychoacoustic model [7] to preserve the perturbative perceptual quality. The resulting adversarial examples can therefore effectively interfere with the encoding process of the speech tokenizer while maintaining high perceptual quality, leading to severe semantic deviations in both ASR and LALM systems.

Furthermore, we aim to explore the reasons behind the effectiveness of this attack and its transferability across different models. Experimental results show that the perturbations are amplified during the conversion from waveforms to Mel-spectrograms, and that the encoder exhibits severe loss of layer-wise information and semantic consistency. As a result, originally small perturbations are progressively amplified by the front end, so that by the time the information reaches the LLM as tokens, it has been completely altered. For example, on the S3 Tokenizer [10], T-SemAttack changes 99.7% of the tokens, which further affects internal processes, e.g., attention disruption in the LALM.

We consider the speech tokenizer as the safety bottleneck in ASR or LALMs. Based on our analysis of the attack, we argue that defense mechanisms should not remain only at the level of the overall model, e.g., safety alignment during post-training, but should also focus on the speech tokenizer. To this end, we propose a Gaussian-noise **ROBUST SpEEch Tokenizer (ROSETok)** that achieves high-fidelity reconstruction while preserving semantically rich representations. Specifically, we utilize deep-layer distillation with a semantic encoder and train the tokenizer with noisy-clean audio pairs to achieve reconstruction and downstream-task robustness against Gaussian noise.

To evaluate the effectiveness of the T-SemAttack, we conduct experiments on bilingual and multi-speaker datasets, 9 ASR systems, and 4 LALM systems, demonstrating that T-SemAttack achieves state-of-the-art (SOTA) performance in terms of perceptual quality, attack efficiency, and effectiveness. In addition, we verify the effectiveness of T-SemAttack against adaptive defenses, including denoising and data transformations. Finally, we experimentally assess the reconstruction and downstream semantic capabilities of the ROSETok under noise conditions. Overall, the contributions of this paper can be summarized as follows:

- We reveal the vulnerability of speech tokenizers and propose a transferable semantic attack method.
- We explore the effectiveness and transferability of our attack from the perspectives of token vulnerability, hierarchical representation drift, and LALM attention interference.
- Motivated by findings of the attack causes, we propose ROSETok, a speech tokenizer that improves noise-robust reconstruction and semantic preservation, enhancing the speech systems’ robustness.
- We conduct extensive experiments to verify the efficiency, performance, and robustness of T-SemAttack. We verify the ROSETok’s performance in reconstruction and downstream scenarios.

2 Background and Related Work

Speech Tokenizer and Speech Foundation Model. A speech tokenizer typically consists of an encoder and quantizer (decoder) [13]. The former encodes speech signals into representations, while the latter discretizes these representations into specific tokens, e.g., VQ- or FSQ-based tokens [10, 14]. As a speech front end, the speech tokenizer plays a crucial role in modern speech foundation models and has been widely adopted in various tasks and systems, including Qwen3-ASR [6] for ASR, CosyVoice [10] for TTS, and Qwen2-Audio [1] for LALMs. Since the speech tokenizer bridges the continuous acoustic space and the discrete semantic space, *safety vulnerability in this component may directly compromise the reliability and safety of the entire speech foundation model.*

Transferable ASR Attacks. Recent adversarial attacks on ASR systems have improved transferability and, in some cases, input-agnostic perturbation generation [8], as summarized in Table 1. However,

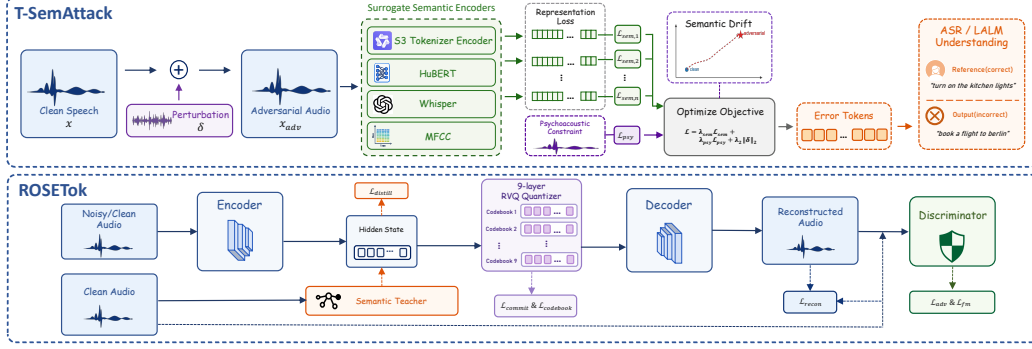


Figure 1: The architecture and pipeline of T-SemAttack and ROSETok.

most existing methods attack the entire ASR pipeline, often relying on surrogate decoding objectives such as CTC loss [7], which can incur high optimization cost and obscure the role of front-end speech representations. Although recent work has observed instability in speech tokenizers [15], tokenizer-level transfer attacks and their propagation mechanisms remain underexplored. In contrast, we focus on the semantic encoder within the speech tokenizer and provide a systematic analysis of how small waveform perturbations induce semantic representation drift, token-level collapse, and downstream LALM attention disruption.

3 Transferable Semantic Attack

Most existing attacks [8, 7, 16] against speech systems rely on specific output targets of ASR or LALM models and require access to the overall architecture, which limits efficiency and transferability. As the front end of LLM-based speech systems, the speech tokenizer plays a crucial role in providing modeling inputs for LLMs, making its security particularly important. We propose T-SemAttack, which attacks semantic representations by jointly perturbing multiple surrogate semantic encoders and uses psychoacoustic constraints to enhance the perceptual quality of the perturbations in Fig. 1.

3.1 Threat Model

Adversary Assumption. We consider a transfer-based attack scenario in which the adversary can access the input speech waveform and possesses a set of differentiable surrogate semantic encoders, but cannot access the internal gradients, parameters, or tokenizer details of the target systems. Under the constraint that the noise remains acceptable, the adversary aims to generate robust adversarial examples that cause speech systems to produce incorrect transcriptions. The adversarial examples should also be robust against adaptive defense, e.g., denoising and data transformations. Moreover, our attack is conducted in an offline setting with minimal real-time requirements.

Defense Assumption. We assume a strong defender who can detect the embedded noise and can use existing relevant tools or defense strategies to restore the original audio as much as possible.

3.2 Problem Definition

Given an original audio signal $x \in [-1, 1]^T$, where T denotes the audio length, our goal is to construct an adversarial example $x_{\text{adv}} = x + \delta$, where δ is a small adversarial perturbation. The perturbation is required to satisfy the following two conditions:

1. *Semantic Disruption:* The x_{adv} should deviate from x in semantic representation spaces, thereby causing discrete token errors with different outputs from ASR and LALMs.
2. *Perceptual Constraint:* The perturbation should remain as acceptable as possible to human listeners, satisfying both amplitude constraints and psychoacoustic constraints.

The overall optimization objective can be written as Eq. (1).

$$\begin{aligned} \min_{\delta} \quad & \mathcal{L} = \lambda_{\text{sem}} \mathcal{L}_{\text{sem}}(x, x_{\text{adv}}) + \lambda_{\text{psy}} \mathcal{L}_{\text{psy}}(x, \delta) + \lambda_2 \|\delta\|_2, \\ \text{s.t.} \quad & \|\delta\|_{\infty} \leq \epsilon, \quad x_{\text{adv}} = x + \delta \in [-1, 1]^T, \end{aligned} \quad (1)$$

where $\mathcal{L}_{\text{sem}}$ measures the deviation of semantic representations, $\mathcal{L}_{\text{psy}}$ represents psychoacoustic masking, and $\|\delta\|_2$ controls perturbation level. We use $\lambda_{\text{sem}} = 100$, $\lambda_{\text{psy}} = 10^{-3}$, and $\lambda_2 = 0.1$.

3.3 Tokenizer-centric Semantic Attack

T-SemAttack. Unlike methods that directly attack ASR decoding outputs [9], our attack is centered on the speech tokenizer. This is because the speech tokenizer is the earliest module in a speech system to perform representation learning and mapping. Once the speech tokenizer is vulnerable to attacks, the subsequent LLM or decoder can hardly recover the original semantic information.

To improve cross-model transferability, we jointly optimize over multiple encoders. Specifically, we construct the following four surrogate branches: (1) **S3 Tokenizer** encoder [10], which directly simulates the semantic encoding process of the speech tokenizer. (2) **HuBERT** [17], which provides deep semantic representations obtained through self-supervised learning. (3) **Whisper** encoder [18], which provides robust semantic representations learned from strong ASR pretraining. (4) Mel-Frequency Cepstral Coefficient (**MFCC**) [12, 7], is used as an additional low-level attack target [7, 9].

Let $\mathcal{F} = f_1, \dots, f_K$ denote the K surrogate semantic encoders. For input x , encoder f_k outputs representations $f_k(x)$. Eq. (2) defines the semantic loss from frame-wise averaged cosine similarity.

$$\mathcal{L}_{\text{sem}} = \sum_{k=1}^K \text{CosSim}(f_k(x), f_k(x_{\text{adv}})). \quad (2)$$

By minimizing Eq. (2), the attack simultaneously drives shifts in multiple semantic spaces, thereby improving transferability to unseen ASR and LALM models.

Mechanistically, this attack can transfer to different ASR and LALM models because multiple semantic encoders, although trained with different objectives, share similar Mel-input bottlenecks and semantic structures. Therefore, disrupting the shared semantic representations in surrogate models often also disrupts the corresponding representations in unseen target models.

Psychoacoustic Model. Previous transferable ASR attacks suffer from low perceptual quality. We utilize a psychoacoustic constraint to ensure that the adversarial perturbation remains acceptable to humans. Given the original audio x , we first compute the corresponding masking threshold $\Theta(x)$ based on its short-time spectrum and use it as an upper-bound reference for the perturbation energy. Let $\text{PSD}(\delta)$ denote the power spectral density of the perturbation in the time-frequency domain. The psychoacoustic loss can then be written as Eq. (3).

$$\mathcal{L}_{\text{psy}} = \sum_{t,f} \max(0, \text{PSD}_{t,f}(\delta) - \Theta_{t,f}(x)), \quad (3)$$

which imposes a penalty only when the perturbation energy in a specific time-frequency unit exceeds the masking threshold. In this way, the optimization process preferentially hides the perturbation in frequency bands or local temporal regions where the original speech is stronger, thereby balancing attack effectiveness and perceptual quality. We follow prior work [19] for the implementation.

Optimization Process. In optimization, we extract multiple representations and update the adversarial audio based on the total loss in Eq. (1). The optimization adopts a PGD-style [20] projected update, where the perturbation is projected back onto the ℓ_∞ ball after each step in Eq. (4).

$$\delta \leftarrow \text{Proj}_{\|\delta\|_\infty \leq \epsilon}(\delta - \alpha \cdot \text{sign}(\nabla_\delta \mathcal{L})), \quad (4)$$

where α is the step size. In practice, we combine the SGD optimizer with an adaptive plateau learning-rate scheduler [7], which automatically reduces the learning rate when the loss stagnates, thereby improving optimization stability. The entire process is run for 500 iterations.

4 Why the T-SemAttack Works?

In Section 7, we conduct extensive experiments demonstrating the effectiveness of T-SemAttack. In this section, we explore why T-SemAttack is effective and how semantic attacks affect the model.

Token Fragility. T-SemAttack disrupts the discretization results of tokenizers. On the LibriTTS dataset [21], T-SemAttack causes extremely high token change rates across multiple tokenizers, reaching 99.702%, 96.666%, 96.418%, and 94.454% on S3 Tokenizer [10], MOSS-Audio-Tokenizer [22],

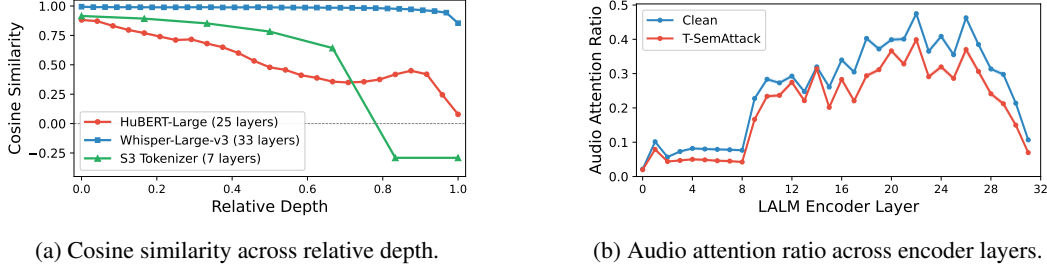


Figure 2: Layer-wise analysis of representation similarity and attention allocation.

156 DAC [23], and StableToken [15], respectively. Semantic attacks disrupt the mapping of speech into
 157 the discrete space, thereby increasing the modeling difficulty for LLMs.

158 **Perturbation Amplification.** On the LibriTTS dataset, we find that T-SemAttack progressively
 159 amplifies perturbations along the waveform ($1\times$) $\rightarrow$ Mel-spectrogram ($5.01\times$) $\rightarrow$ encoder ($9.87\times$)
 160 chain when applied to S3 Tokenizer. Details are provided in Appendix G.1.

161 **Layer-wise Similarity.** Experiments on LibriTTS show that different model layers encode different
 162 types of information, as shown in Fig. 2a. For HuBERT [17], the cosine similarity between clean and
 163 adversarial representations decreases from 0.881 at the input embedding layer to 0.079 at the final
 164 layer. Although Whisper [18] remains relatively stable in the early layers, its final-layer similarity still
 165 drops to 0.854. For S3 Tokenizer [10], high-level representations even become negatively correlated
 166 (-0.291). These results indicate that shallow acoustic representations are relatively robust, whereas
 167 perturbations are more likely to accumulate and be amplified by the Transformer architecture as they
 168 approach the semantic layers, making lightweight tokenizers particularly vulnerable.

169 **LALM Attention Disruption.** Incorrect tokens further disrupt cross-modal modeling in LALMs.
 170 Attention analysis of Qwen2-Audio [1] shows that in the higher Transformer layers (layers 9–28), the
 171 attention ratio from the target text to audio tokens decreases by approximately 15%–25% on average,
 172 *while the reduced attention is mainly shifted to the target text itself*, as shown in Fig. 2b. This further
 173 interferes with the model’s understanding. Detailed visualization is shown in Appendix G.2.

174 5 Mechanism-Guided Robust Speech Tokenizer

175 In Section 4, we investigate why T-SemAttack can achieve effective attacks and reveal an important
 176 insight: some existing speech tokenizers are not semantically robust and are vulnerable to adversarial
 177 attacks. We believe that defense solely at the LLM level is more time- and cost-intensive. Therefore,
 178 we return to the core of the problem and aim to improve the stability of the tokenizer against noise.
 179 Analysis shows that perturbations can be amplified in the Mel-spectrogram domain, and deep semantic
 180 information drifts. Thus, we expect the tokenizer to internally satisfy two requirements: noisy training
 181 and semantic preservation. In this work, we propose a Noise-Robust Speech Tokenizer (ROSETok),
 182 which can reconstruct high-quality audio from noisy inputs.

183 **Model Architecture.** We design a convolutional architecture based on DAC [23]. The encoder
 184 takes a waveform as input and compresses it $320\times$, producing a 50 Hz representation. The quantizer
 185 consists of 9 RVQ layers. The decoder mirrors the encoder. Details are provided in Appendix F.3.

186 **Noisy Training.** As shown in Fig. 1, we use noisy audio as input and clean audio as supervision for
 187 both reconstruction and semantic signals during training, thereby enabling noise-robust, high-quality
 188 audio reconstruction. The training input consists of 20% clean audio and 80% noisy audio.

189 **Semantic Preservation Module.** Fig. 2a illustrates the severe loss of deep semantic information. We
 190 employ a pretrained Semantic Teacher, Whisper [18], to supervise the representations of the model
 191 encoder. Specifically, we align the representations of noisy inputs with those of clean audio, thereby
 192 preserving semantic information and improving robustness to noise in downstream tasks.

193 **Training.** We aim for high-fidelity reconstruction and semantic consistency. For both clean and noisy
 194 inputs, the model outputs high-quality clean audio. Moreover, we hope the encoder can preserve
 195 semantically rich information even for noisy audio inputs. We use a multi-frequency discriminator

with hinge loss [24] for adversarial learning. The objective is in Eq. (5).

$$\mathcal{L}_{\text{tok}} = \lambda_{\text{distill}}\mathcal{L}_{\text{distill}} + \lambda_{\text{recon}}\mathcal{L}_{\text{recon}} + \lambda_{\text{cb}}\mathcal{L}_{\text{cb}} + \lambda_{\text{commit}}\mathcal{L}_{\text{commit}} + \mathcal{L}_{\text{adv}} + \mathcal{L}_{\text{fm}}, \quad (5)$$

where $\mathcal{L}_{\text{distill}}$ is the semantic distillation loss, $\mathcal{L}_{\text{recon}}$ is the reconstruction loss, $\mathcal{L}_{\text{cb}}$ is the code-book loss, $\mathcal{L}_{\text{commit}}$ is the commitment loss, $\mathcal{L}_{\text{adv}}$ is the adversarial loss, and $\mathcal{L}_{\text{fm}}$ is the feature matching loss. Detailed descriptions are provided in Appendix F.3. The hyperparameters are set as $\{\lambda_{\text{distill}}, \lambda_{\text{recon}}, \lambda_{\text{cb}}, \lambda_{\text{commit}}\} = \{5.0, 15, 1.0, 0.25\}$. The weights of $\mathcal{L}_{\text{adv}}$ and $\mathcal{L}_{\text{fm}}$ are both 1.

6 Experimental Setup

6.1 Semantic Attack Setup

Datasets. To comprehensively evaluate the effectiveness of the semantic attack, we use bilingual and multi-speaker speech datasets. LibriTTS [21]: Following [7], we select single-speaker samples to evaluate the effectiveness of the attack in an English single-speaker scenario. SeedTTS-Eval [25]: This is a widely used benchmark, containing both Chinese and English utterances from multiple speakers. We randomly sample 200 English utterances and 200 Chinese utterances for evaluation.

ASR Systems. We evaluate the attack performance on both open-source models (Qwen3-ASR [6], VibeVoice-ASR [26], FireRed-ASR [27], and Fun-ASR [28]) and commercial black-box models (iFlytek, Doubao ASR, and GPT-4o-mini-transcribe).

LALM Systems. The safety of LALMs is also a major focus of this work. We further evaluate our attack on several representative LALMs, including the open-source models Qwen2-Audio [1], Qwen3-Omni [2], and Audio-Flamingo 3 [5], as well as the black-box model Gemini-3-Flash.

Metrics. For efficiency, we report the Real-Time Factor (RTF) to measure the computational cost of the attack. For perceptual quality, we use Signal-to-Noise Ratio (SNR), Perceptual Evaluation of Speech Quality (PESQ), and Speaker Similarity (SIM) [29] to evaluate the degree of distortion introduced by the attack in terms of audio quality and speaker consistency. For attack effectiveness, we transcribe the adversarial audio using the target model and compute the Word Error Rate (WER). A higher WER indicates more severe corruption of the semantic content in the original speech.

6.2 Training Details of the Robust Tokenizer

Datasets. We train the robust tokenizer on approximately 96.7k hours of bilingual speech data in Chinese and English. We utilize the English and Chinese subset from the Emilia [30] dataset.

Training Hyperparameters. We train the model on 8 NVIDIA H100 GPUs with a global batch size of 128. We use the AdamW optimizer, with an initial learning rate of 1×10^{-4} and $\beta_1 = 0.8, \beta_2 = 0.99$. The model is trained for a total of 200k steps, and the learning rate is decayed by 10%.

7 Experiments

In this section, we evaluate T-SemAttack and ROSETok. First, we verify the efficiency and perceptual quality of the attack, and verify the effectiveness and transferability of semantic attacks on ASR systems. Second, we evaluate the impact of the attack on cross-modal audio understanding in LALMs. Third, we test the adaptive robustness of the attack against denoising, adversarial defense, and audio transformation techniques. For ROSETok, we evaluate its reconstruction stability on both clean and noisy audio. Finally, we conduct ablation studies for validation.

7.1 Evaluation on ASR Systems

Table 2 reports the main results. T-SemAttack achieves a better balance among efficiency, perceptual quality, and attack strength. Its RTF is 21.800, lower than E2E-VGuard’s 26.797, while maintaining an SNR of 16.319 with good perceptual quality. Subjective experiments are provided in Appendix E. For efficiency comparison of AudioShield, details and results are provided in Appendix A and F.5. AudioShield can produce high WERs when $\tau = 0.3$ (where τ is a hyperparameter), but it almost destroys the audio quality, making the content difficult to hear clearly. It also suffers from noticeable timbre loss when $\tau = 0.3/0.1$. Therefore, in the following experiments, we use $\tau = 0.1$.

Table 2: Comparison of attack efficiency, perceptual quality, and attack effectiveness across ASR systems and datasets. WER (%) is reported, where higher values indicate stronger attack effectiveness. The best and second-best results are highlighted with **bold** and underlined, respectively.

Methods	Efficiency	Perception			Surrogate Model		Open-source Model			Black-box Model			Avg.	
	RTF(↓)	SNR(↑)	PESQ(↑)	SIM(↑)	HuBERT	Whisper	Qwen3-ASR	VibeVoice	FireRed-ASR	Fun-ASR	iFlytek	Doubao		GPT-4o
LibriTTS: A single-speaker English dataset.														
Clean	-	-	-	-	2.694	2.079	2.369	2.156	2.017	1.866	3.648	3.463	2.234	2.503
Gaussian Noise	0.003	15.197	2.332	0.864	3.228	2.213	2.583	2.736	2.933	2.488	3.837	3.691	6.677	3.376
E2E-VGuard [7]	26.797	14.341	2.085	0.756	10.781	5.834	6.781	8.325	11.376	8.842	8.672	7.618	8.030	8.473
AudioShield-0.3 [8]	-	-2.862	1.137	0.294	60.083	58.702	63.429	72.605	59.379	67.312	64.817	63.192	55.546	62.785
AudioShield-0.1 [8]	-	-2.440	1.178	0.361	16.007	10.363	15.450	15.241	13.800	12.171	16.815	14.273	11.049	13.908
T-SemAttack	21.800	16.319	2.055	0.445	156.913	111.367	<u>27.388</u>	<u>63.555</u>	<u>33.529</u>	<u>43.679</u>	<u>49.607</u>	<u>20.004</u>	<u>46.089</u>	<u>61.348</u>
Seed-TTS (EN): A multi-speaker English dataset.														
Clean	-	-	-	-	6.286	0.851	1.040	1.433	5.906	1.015	1.942	3.001	1.907	2.598
Gaussian Noise	0.003	15.228	2.249	0.830	10.739	1.307	1.695	3.862	10.985	2.061	3.457	2.679	9.849	5.182
E2E-VGuard [7]	26.797	12.907	1.268	0.677	42.084	15.913	15.643	26.609	35.936	20.541	19.051	14.233	18.374	23.154
AudioShield-0.1 [8]	-	-2.364	1.235	0.212	<u>62.779</u>	<u>42.840</u>	50.674	<u>51.296</u>	<u>54.301</u>	<u>56.806</u>	<u>54.181</u>	61.297	<u>42.172</u>	<u>52.927</u>
T-SemAttack	21.800	17.598	2.119	0.423	189.289	125.913	<u>37.445</u>	103.887	65.406	63.779	71.726	29.948	49.457	81.872
Seed-TTS (ZH): A multi-speaker Chinese dataset.														
Clean	-	-	-	-	-	3.086	0.790	1.521	1.760	0.614	1.560	1.109	2.984	1.678
Gaussian Noise	0.003	15.387	1.977	0.820	-	5.434	1.207	6.171	2.899	1.466	1.331	0.774	15.129	4.301
E2E-VGuard [7]	26.797	<u>13.607</u>	1.719	0.634	-	18.392	5.322	27.258	8.896	6.879	3.119	1.944	18.279	11.261
AudioShield-0.1 [8]	-	-2.435	1.181	0.424	-	<u>28.113</u>	19.998	<u>50.729</u>	16.609	<u>23.267</u>	<u>14.345</u>	30.221	<u>37.687</u>	<u>27.621</u>
T-SemAttack	21.800	17.652	1.886	0.436	-	109.100	<u>18.833</u>	144.637	19.293	29.021	19.214	6.298	60.116	50.814

Table 3: Understanding performance comparison across different datasets and LALMs by WER(%).

Method	LibriTTS				Seed-TTS (EN)				Seed-TTS (ZH)		
	Qwen2-A	Qwen3-O	Audio-F3	Gemini3	Qwen2-A	Qwen3-O	Audio-F3	Gemini3	Qwen2-A	Qwen3-O	Gemini3
Clean	2.982	1.645	4.428	3.345	4.533	5.060	5.611	0.842	5.191	0.455	2.391
Gaussian Noise	3.831	1.787	5.920	2.530	4.184	2.020	10.389	1.841	2.762	2.593	5.619
E2E-VGuard	11.833	3.907	11.782	35.284	22.428	11.872	36.651	15.915	10.146	4.616	19.885
AudioShield	14.762	7.600	15.556	19.501	66.674	45.825	65.748	70.153	32.797	18.642	68.023
T-SemAttack	97.280	18.646	37.944	55.855	102.486	29.777	56.590	68.493	79.259	13.687	81.300

In contrast, under better perceptual quality, T-SemAttack increases the WERs of multiple open-source and black-box ASR systems from about 1%~4% on clean speech to 20.004%~63.555%, reaching 49.607% and 46.089% on iFlytek and GPT-4o-mini-transcribe, respectively.

Table 2 verifies the generalization ability in multi-speaker and bilingual scenarios. On Seed-TTS (EN), T-SemAttack achieves an average WER of 81.872% on ASR systems, significantly higher than E2E-VGuard’s 23.154%. On the Chinese subset, the average WER remains 50.814%. Notably, T-SemAttack achieves a cross-linguistic attack on the Chinese dataset. This is because our surrogate model, Whisper, and S3 Tokenizer are trained on a large scale of bilingual datasets. Notably, commercial systems show larger robustness differences in Chinese scenarios. This experiment indicates that the transferability of semantic attacks generally holds.

7.2 Evaluation on LALMs

Table 3 shows the impact of T-SemAttack on the speech understanding ability of LALMs. Compared with ASR systems, LALMs not only need to recognize speech content, but also *align audio tokens with the text reasoning process*. Therefore, they are more sensitive to semantic drift at the tokenizer level. T-SemAttack achieves an average WER of 58.30% over 11 dataset-model combinations, which is much higher than E2E-VGuard’s 16.76% and AudioShield’s 38.66%. For example, on LibriTTS, the WER of Qwen2-Audio increases from 2.982% to 97.280%. On the Chinese set, Qwen2-Audio and Gemini3-Flash also reach 79.259% and 81.300%, respectively. These results show that once the semantic tokenizer produces incorrect semantic tokens, subsequent LALMs can hardly recover the original semantics through stronger language modeling.

This phenomenon is also consistent with the attention analysis in Section 4 which shows that attacked audio tokens weaken the model’s reliance on audio context, causing cross-modal attention to shift more toward the text itself and thereby amplifying semantic understanding errors.

7.3 Adaptive Defense against Semantic Attack

We consider some strong defense settings: denoising, adversarial defense, diffusion-based adversarial defense, and audio transformations. This experiment is conducted on the SeedTTS-EN dataset.

Table 4: Adaptive defense against advanced denoising models across three models, adversarial audio defense, and audio transformations on the VibeVoice-ASR with averaged WER(%) values.

Method	Denoiser [31]			DeepFilterNet3 [32]			Adversarial Defense [33]			Diffusion [34]	Audio Transformation [7]			
	HuBERT	Whisper	VibeVoice	HuBERT	Whisper	VibeVoice	Resample	Mel	Q-D	AudioPure	Speed	MP3	PS	Band-filter
Gaussian Noise	3.414	2.964	1.720	2.948	2.468	1.133	5.256	6.569	2.984	8.737	57.517	3.653	3.735	20.334
E2E-VGuard	13.042	10.205	7.276	11.762	10.168	7.148	13.841	17.650	9.447	13.898	67.112	10.586	10.282	28.863
AudioShield	16.255	9.930	9.738	16.075	10.741	9.187	16.030	16.751	14.673	37.497	61.546	15.357	13.379	23.901
T-SemAttack	88.258	96.763	42.901	83.185	113.997	37.805	55.805	58.997	73.364	21.571	89.898	66.189	70.635	71.085

Table 5: Reconstruction evaluation on different speech tokenizers under clean and noisy inputs.

Model	N_q	Frame Rate	RTF($\downarrow$)	Clean Input				Noisy Input			
				Seed-TTS (EN)		Seed-TTS (ZH)		Seed-TTS (EN)		Seed-TTS (ZH)	
				STOI($\uparrow$)	PESQ($\uparrow$)	STOI($\uparrow$)	PESQ($\uparrow$)	STOI($\uparrow$)	PESQ($\uparrow$)	STOI($\uparrow$)	PESQ($\uparrow$)
				<i>Acoustic and Semantic Speech Tokenizer</i>							
SNAC [35]	4	–	0.003	0.872	1.847	0.867	1.894	0.812	1.318	0.780	1.372
XY-Tokenizer [36]	8	12.5	0.025	0.901	2.198	0.909	2.362	0.820	1.324	0.784	1.334
XCodec-2 [24]	1	50	0.021	0.899	2.113	0.901	2.250	0.821	1.354	0.785	1.366
DAC [23]	9	50	0.008	0.943	3.168	0.939	3.172	0.800	1.390	0.834	1.623
<i>Robust Speech Tokenizer</i>											
MOSS [22]	9	12.5	0.029	0.931	2.540	0.936	2.727	0.886	1.494	0.867	1.529
StableToken [15]	1	25	0.175	0.647	1.043	0.662	1.061	0.643	1.042	0.660	1.059
ROSETok (Ours)	9	50	0.010	0.965	3.063	0.962	3.085	0.950	2.431	0.938	2.380

Advanced Denoising. We utilize two SOTA denoising models, Denoiser [31] and DeepFilterNet3 [32]. The results in Table 4 show that after Denoiser processing, T-SemAttack still achieves WERs of 88.258%, 96.763%, and 42.901% on HuBERT, Whisper, and VibeVoice-ASR, respectively. The WERs remain high after DeepFilterNet3 processing. In contrast, under the same defenses, the WERs of E2E-VGuard and AudioShield are mostly only 7%~16%.

Adversarial Defense and Audio Transformations. We consider adversarial defense based on WaveGuard, diffusion-based defense (AudioPure [34]), and audio transformation-based defense. The results are shown in Table 4. The best defense method is AudioPure, achieving a WER score of 21.571%, and still misleads correct transcription. The highest WER of AudioPure is 91.667%, reflecting that this method cannot perfectly disrupt the T-SemAttack. We find that previous methods are not robust enough against these advanced defensive methods, and T-SemAttack achieves SOTA performance. The explanations of these techniques are provided in the Appendix F.2.

These results suggest that T-SemAttack perturbations are not merely removable noise by conventional speech enhancement. Although Denoiser and DeepFilterNet3 can suppress the acoustic disturbance, they cannot reliably recover the original token sequence or semantic representation once the tokenizer output has been shifted. This indicates that speech enhancement alone is insufficient, and *ROSETok must explicitly model both noise robustness and semantic consistency at the tokenizer level.*

7.4 Robust Speech Tokenizer Analysis

Robust Speech Reconstruction. In this section, we compare ROSETok with existing speech tokenizers in terms of reconstruction quality under clean and noisy inputs. Experimental results are shown in Table 5. For clean inputs, ROSETok achieves STOI scores of 0.965/0.962 and PESQ scores of 3.063/3.085 on Seed-TTS (EN/ZH), respectively. Its overall performance is close to MOSS-Audio-Tokenizer [22], while maintaining a low RTF of 0.010. This indicates that its robust design does not significantly sacrifice clean-speech reconstruction ability. For noisy inputs with noise levels ranging from 2/255 to 16/255, ROSETok shows a more obvious advantage that its STOI reaches 0.950/0.938, and its PESQ reaches 2.431/2.380, substantially higher than MOSS-Audio-Tokenizer’s 0.886/0.867 and 1.494/1.529, and also clearly better than models, e.g., XY-Tokenizer and XCodec-2.

Various Noise-Level Reconstruction. We evaluate the noise robustness of models under Gaussian noise with magnitudes ranging from 1/255 to 24/255 with results shown in Fig. 3. Most models suffer severe performance degradation under stronger noise, e.g., DAC drops from 0.943 on clean input to 0.770 at the 24/255 noise level. In contrast, ROSETok remains relatively stable and still achieves good reconstruction quality under strong noise (24/255), with a 0.920 STOI score.

ROSETok improves robustness to generic Gaussian perturbations, but it should not be interpreted as a complete mitigation against structured semantic attacks, e.g., T-SemAttack. Attack-aware fine-tuning

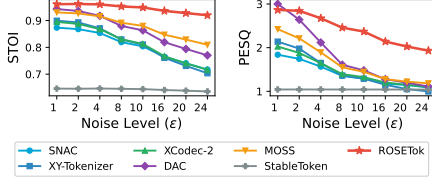


Figure 3: Noise-level evaluation.

Table 6: Ablation results on reconstruction and semantic evaluation of the FSC task. “w/o” represents without this loss/component.

Method	Recon-Clean		Recon-Noisey		Semantic-FSC(↑)	
	STOI(↑)	PESQ(↑)	STOI(↑)	PESQ(↑)	Clean	Noisy
w/o noisy	0.956	2.892	0.879	1.578	69.259	11.363
w/o $\mathcal{L}_{\text{distill}}$	0.947	2.652	0.934	2.322	7.118	5.220
Ours	0.945	2.573	0.931	2.304	61.139	60.032

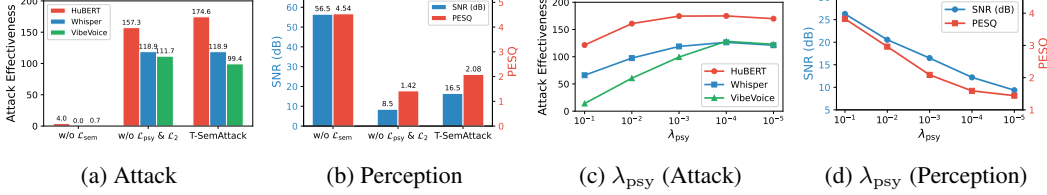


Figure 4: Ablation study for the T-SemAttack. (a)(b) Component Effect. (c)(d) Sensitivity to λ_{psy} .

301 further improves robustness, suggesting that robust tokenizers require both generic noise robustness
 302 and attack-aware semantic augmentation. Details are introduced in Appendix A.

303 7.5 Ablation Study

304 In this section, we conduct ablation experiments. For the attack, we randomly select 30 samples from
 305 Seed-TTS (EN). For ROSETok, we train on the LibriSpeech [37] for 100k steps.

306 **Component Effect of T-SemAttack.** We study the effect of each component in Eq. (1) of T-
 307 SemAttack. Results are shown in Fig. 4a and 4b. We find that, after removing the semantic attack loss
 308 $\mathcal{L}_{\text{sem}}$, the perceptual quality remains high, but the attack is almost ineffective on HuBERT, Whisper,
 309 and VibeVoice-ASR, with an attack success rate of only about 1.556%. After removing the perceptual
 310 loss $\mathcal{L}_{\text{psy}}$ and $\mathcal{L}_2$, the SNR drops to 8.5, and the PESQ score falls below 2.08. Combining both
 311 components achieves a sound balance between perceptual quality and attack effectiveness.

312 **Perception and Attack Balance.** In Eq. (1), λ_{psy} balances attack effectiveness and perceptual
 313 quality. We evaluate its sensitivity over the range from 10^{-1} to 10^{-5} . Results in Fig. 4c4d show that
 314 when λ_{psy} is set to 10^{-3} or smaller, the attack effectiveness on the three models shows no significant
 315 improvement, while the SNR continues to decrease. Therefore, we set the λ_{psy} value to 10^{-3} .

316 **Component and Semantic Effect of ROSETok.** We verify the effect of ROSETok’s components at
 317 noise levels of $2/255 \sim 16/255$. As shown in Table 6, removing noisy training reduces noisy-input
 318 STOI from 0.931 to 0.879 and PESQ from 2.304 to 1.578, confirming the importance of noisy-clean
 319 training. To evaluate semantic capability and consistency, we adapt the ROSETok encoder to the
 320 XARES [38] benchmark. Higher benchmark scores are better. On Fluent Speech Commands (FSC),
 321 a small-scale ASR task, we train on the clean training set and evaluate on its clean and noisy test sets
 322 to assess the model’s robustness to noise in downstream scenarios. When the noisy signal input is
 323 removed, the FSC score drops from 69.259 to 11.363. Removing semantic supervision leads to even
 324 more severe semantic degradation on FSC. Overall, ROSETok achieves high-fidelity reconstruction
 325 of both clean and noisy audio while preserving strong semantic robustness under noisy conditions.

326 8 Conclusion

327 In this work, we explore the robustness of speech tokenizers. Based on the semantic encoder, we
 328 propose T-SemAttack, which effectively attacks current mainstream ASR and LALM systems. We
 329 further analyze the causes of the attack and, inspired by these findings, propose a robust speech
 330 tokenizer that enables high-quality reconstruction from noisy speech. These findings highlight the
 331 speech tokenizer as a critical safety bottleneck in modern speech foundation models. We hope our
 332 analysis and ROSETok can provide useful insights for developing future speech tokenizers with
 333 stronger semantic consistency and robustness.

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468 A Discussion and Limitations

469 In this section, we discuss some important aspects.

470 Q1: Can T-SemAttack’s perturbation be really imperceptible to humans?

471 **A:** In the setting considered in this paper, what we aim to emphasize is that the noise does not
472 compromise the normal usability of the original audio, that is, its **harmlessness**, rather than true
473 imperceptibility. Compared with the baselines, we find that when the value of τ in AudioShield is
474 0.3, human listeners can barely perceive the content of the original audio. When τ is 0.1, the timbre
475 information of some audio samples is also altered. The reason is that AudioShield is trained on
476 LibriSpeech and has learned a large amount of female voice characteristics. Moreover, modifying
477 the latent representation cannot guarantee the quality of the reconstructed original waveform. T-
478 SemAttack adopts a linear additive strategy and incorporates the masking effect to minimize its
479 impact on the original audio content as much as possible. Subjective experiments in Appendix E
480 show that T-SemAttack achieves better audio quality, content consistency, and speaker similarity
481 than baselines, achieving acceptable and normal use of the attacked audio. In addition, T-SemAttack
482 exhibits strong robustness, which ensures that even if the defender perceives the embedded noise,
483 denoising methods, adversarial defenses, and data transformations cannot effectively remove or
484 neutralize it. *Therefore, T-SemAttack aims to avoid disrupting the normal use of audio, rather than*
485 *to achieve true imperceptibility.*

486 Q2: How efficient is T-SemAttack?

487 **A:** First, we focus on an offline scenario, where the real-time requirement is relatively low. In Table
488 2, we evaluate the RTF of our method and the baselines. Specifically, the RTF is calculated as the
489 total time required by each method to attack the same LibriTTS dataset divided by the total duration
490 of the audio. Details of efficiency computation are provided in Appendix F.5.

491 T-SemAttack achieves an RTF of 21.8, outperforming E2E-VGuard of 26.797. This is because our
492 attack targets the semantic encoder rather than the entire model, which partially improves efficiency.
493 It is worth noting that AudioShield is a real-time attack framework, but it requires a very long
494 initialization stage. Specifically, before AudioShield can be used to attack audio, it must first be
495 trained on LibriSpeech for an extended period of time. Therefore, the RTF of AudioShield is 87.045
496 for warmup-based training and 0.040 for each sample attack after long-time offline training.

497 Therefore, although T-SemAttack currently achieves better time efficiency than E2E-VGuard, improv-
498 ing its real-time performance remains a direction for future work, e.g., multi-GPU parallelization.

499 Q3: Can ROSETok resist T-SemAttack?

500 **A:** In Section 4, we analyze why T-SemAttack is effective. Based on the discovered chain of pertur-
501 bation amplification, we propose ROSETok, a mechanism-driven noise-robust model architecture.
502 Because our training pairs consist of Gaussian-noised and clean audio, ROSETok also demonstrates
503 robustness and semantic preservation under Gaussian-noise input scenarios in Section 7.

504 On unseen samples attacked by T-SemAttack, however, ROSETok exhibits degraded reconstruction
505 quality, achieving an STOI score of 0.842 on SeedTTS-EN and 0.787 on the Chinese set. This
506 performance drop is expected because ROSETok is trained with Gaussian-noisy/clean audio pairs
507 and is not exposed to structured semantic perturbations generated by T-SemAttack. Rather than
508 contradicting the design of ROSETok, this result reveals a distributional gap between generic noise
509 robustness and attack-aware semantic robustness: Gaussian noise mainly tests local acoustic stability,
510 whereas T-SemAttack induces coordinated shifts in tokenizer-level semantic representations.

511 This observation suggests that robust speech tokenizers should not rely solely on conventional noisy
512 training. Instead, they should incorporate attack-aware semantic augmentation so that the tokenizer
513 can learn to align perturbed semantic trajectories with clean speech representations. To test this, after
514 fully pretraining ROSETok for 200k steps on a large-scale dataset, we fine-tune it using only 200
515 samples attacked by T-SemAttack for 2,000 steps on a single GPU. On the Chinese dataset, the STOI
516 score improves from 0.787 to 0.888, substantially reducing the degradation caused by T-SemAttack.

517 These results indicate that ROSETok provides a useful mechanism-guided model for robust speech
518 tokenization, but its robustness depends on the coverage of the training distribution. Therefore,
519 ROSETok should not be interpreted as a complete out-of-the-box mitigation against T-SemAttack.

520 Instead, it demonstrates a practical defense direction: pretraining tokenizers with generic noisy-clean
521 reconstruction and semantic distillation, followed by lightweight attack-aware adaptation when
522 stronger semantic attacks are identified.

523 **Q4: Why does ROSETok preserve semantics at the encoder-level rather than the token-level?**

524 **A:** Token-level stability and encoder-level semantic preservation address different failure modes.
525 Token consistency encourages discrete-code invariance, while ROSETok targets the semantic rep-
526 resentation drift identified by T-SemAttack. Since token changes do not always imply semantic
527 degradation, we align encoder representations with clean semantic teachers rather than enforcing
528 exact token invariance. Specifically, we consider two reasons.

529 First, this approach has broader applicability. Some recent codecs [39] introduce semantic distillation
530 in the first few layers of RVQ and use the subsequent layers to model acoustic information. Other
531 methods [15] directly constrain the stability of quantized tokens under noisy inputs. In contrast,
532 ROSETok directly constrains the semantic representations output by the encoder, encouraging them
533 to remain close to the semantic representations of clean speech under noisy inputs. This not only
534 provides more robust inputs for the subsequent quantizer, but also makes the approach applicable to
535 recent speech tokenizers [40] that are based on continuous representations rather than discrete tokens.
536 Accordingly, ROSETok is designed precisely to align the semantic representations of noisy inputs
537 with those of clean audio using a Whisper teacher.

538 Second, changes in tokens do not necessarily imply changes in semantics. Random Gaussian noise
539 may cause a large proportion of the tokenizer’s output tokens to change. However, if the deep
540 semantic representations remain stable, downstream ASR or LALM models may still recover the
541 correct semantics through their own modeling capabilities. By contrast, the key mechanism of T-
542 SemAttack is not simply to change token IDs, but to drive semantic drift in the encoder representations,
543 making it difficult for downstream models to recover the original content.

544 Therefore, the goal of ROSETok is not to keep tokens unchanged, but to align the semantic repre-
545 sentations of perturbed inputs with those of clean speech. Compared with token-level consistency,
546 encoder-level semantic preservation corresponds more directly to the attack mechanism identified in
547 this paper and, in this work, serves as a more suitable training objective for robust speech tokenizers.

548 **B Ethical Considerations and Broader Impacts**

549 **Dual-Use and Disclaimer of the T-SemAttack.** We recognize that T-SemAttack is a dual-use
550 technique. While this work exposes security threats in existing speech tokenizers and encourages
551 researchers to develop more robust speech tokenizers, the same technique could be misused to
552 degrade automatic transcription, evade speech-based moderation or auditing systems, interfere with
553 voice assistants, or reduce the reliability of speech interfaces used in accessibility, documentation, or
554 security-sensitive applications. We therefore treat the proposed attack as a security research artifact
555 rather than a deployable tool. To mitigate unforeseen impacts, before releasing the source code, we
556 will publish a disclaimer stating that our tool is intended solely for academic research and related
557 purposes. Users must not use this tool for unauthorized or illegal activities, and all actions taken by
558 users are unrelated to the creators and publishers of the tool.

559 **Use of Third-Party Systems.** Our black-box evaluation uses commercial ASR and LALM APIs only
560 to assess transferability. We do not attempt to bypass authentication, rate limits, monitoring, or any
561 access-control mechanism. The evaluation is conducted at a limited scale, uses only public benchmark
562 utterances, and does not involve private user speech. We report aggregate performance metrics and
563 do not use the services to extract model parameters, training data, or proprietary information.

564 **Subjective Experiments.** We obtain informed consent from all participants during recruitment, and
565 all selected participants are over 18 years old. The identities of the participants are anonymized.
566 We do not collect any information beyond what is necessary for the subjective experiments, and
567 the relevant information and responses are used only for academic research. We will not disclose
568 any such sensitive information. All adversarial audio samples are discarded after the subjective
569 experiments are completed, so they will not cause any additional commercial or societal impact.

570 **Defensive Intent and Limitations.** We also release and evaluate ROSETok to encourage defensive
571 research on robust speech tokenizers. Nevertheless, ROSETok should not be interpreted as a complete

mitigation against all semantic attacks. Its robustness depends on the training distribution and threat model, and deployed systems should combine tokenizer-level robustness with additional monitoring, denoising, adversarial evaluation, and system-level safeguards.

C Case Study: T-SemAttack in Text-To-Speech for Protection

To evaluate the potential application of T-SemAttack in speech protection scenarios, we conduct a case study on text-to-speech (TTS) systems. Specifically, we consider a realistic threat in which an attacker can obtain a user’s speech samples and utilize advanced TTS systems for voice cloning [7]. If the original speech is slightly perturbed before being released or uploaded in a form that remains acceptable to humans, downstream speech generation systems would struggle to accurately extract speaker characteristics or semantic information. Motivated by this scenario, we evaluate whether T-SemAttack can serve as a speech protection against the unauthorized use of TTS systems.

We select three representative TTS systems for evaluation, i.e., CosyVoice [10], CosyVoice2 [14], and Step-Audio-TTS [41]. We evaluate the generated speech from two perspectives: WER and SIM. A higher WER indicates more severe semantic deviation in the generated content, while a lower SIM indicates that the system has greater difficulty preserving the voice characteristics of the original speaker. We conduct experiments on the Seed-TTS (EN) dataset, using the ground-truth transcription of each audio sample as the input reference text.

Table 7: Case study of T-SemAttack in the TTS protection scenario. The higher WER(%) represents more severe semantic errors in the generated speech. The lower the SIM, the lower the similarity between the generated speech and the original speaker.

Method	CosyVoice [10]		CosyVoice2 [14]		Step-Audio-TTS [41]	
	WER(↑)	SIM(↓)	WER(↑)	SIM(↓)	WER(↑)	SIM(↓)
Clean	4.583	0.599	4.484	0.836	3.734	0.772
T-SemAttack	52.476	0.212	33.427	0.330	122.746	0.299

Table 7 presents the results of this case study. With clean input, all three TTS systems preserve the semantic content well, achieving WERs of only 4.583, 4.484, and 3.734, respectively. These observations suggest that these systems can effectively use the speaker features contained in the input speech for speech generation. In contrast, when the input speech is processed by T-SemAttack, the generation quality of all TTS systems degrades substantially. Step-Audio-TTS is affected most severely, with its WER increasing sharply from 3.734 to 122.746. On average, the WER increases from 4.267 under the clean condition to 69.550, while the speaker similarity decreases from 0.736 to 0.280. These results show that T-SemAttack can disrupt the modeling of speech content in TTS.

This result suggests that semantic-level perturbations can also affect the conditional modeling process in TTS systems. When a TTS system relies on input speech to extract semantic or speaker representations, the semantic representation disruption and tokenizer-level representation perturbation caused by T-SemAttack can affect the generation stage, resulting in content errors and reduced speaker similarity in the generated speech. Therefore, this case study provides another perspective supporting the critical role of speech tokenizers in modern speech systems.

D Robustness and Semantic Analysis of ROSETok

Fig. 3 and Table 6 have demonstrated the robustness of ROSETok. In this section, we further supplement the experiments on the semantic evaluation of ROSETok.

When designing ROSETok, we distill the encoder representations of noisy audio inputs and the semantic representations of clean audio inputs, thereby ensuring semantic preservation under both clean and noisy conditions. We first evaluate the semantic ability of the ROSETok encoder by feeding the 50 Hz representations into the XARES benchmark and testing them on multiple speech tasks.

Table 8 shows the results of semantic evaluation on downstream tasks. In the semantic evaluation, we find that after distillation with the semantic encoder, ROSETok preserves as much semantic information compared to pretrained Self-Supervised Learning (SSL) models, e.g., WavLM [42]

Table 8: Semantic and downstream evaluation on classification and ASR tasks.

Model	Semantic Capability			ASR-FSC		
	CREMAD($\uparrow$)	LibriCnt($\uparrow$)	LSMF($\uparrow$)	Clean	Noisy	Δ
WavLM [42]	72.133	64.965	97.165	97.812	64.487	$\downarrow 33.325$
Wav2Vec2 [43]	55.735	58.794	93.958	48.932	11.152	$\downarrow 37.780$
ROSETok	56.362	61.661	96.565	73.899	72.027	$\downarrow 1.872$

and Wav2Vec2 [43]. In addition, to evaluate robustness on downstream tasks, we test the FSC task under noisy test-set conditions, with noise magnitudes ranging from 2/255 to 16/255. We find that pretrained semantic encoders largely degrade downstream performance under noisy conditions, such as 48.932 $\rightarrow$ 11.152 on the FSC task of the Wav2Vec2 model. Compared to these models, we observe that on the FSC task, ROSETok achieves 72.027 on the noisy test set with only 1.872 degradation (Δ : the score of the clean value minus the noisy value), which is comparable to that obtained on the clean test set. This further demonstrates the robustness of ROSETok to noise in downstream tasks.

The abbreviated tasks are as follows: FSC denotes Fluent Speech Commands; CREMA-D denotes the Crowd-sourced Emotional Multimodal Actors Dataset; LibriCnt denotes speaker count; and LSMF denotes LibriSpeech Male and Female.

E Human Study

To evaluate the human perception of the perturbation, we conduct a subjective human listening study to assess the perceptual quality of the attacked audio generated by T-SemAttack and the baselines. This section details the recruitment process, design of the questionnaire, and results of evaluations.

Recruitment Process. This subjective survey has been approved by the Human Ethics Research Committee at the first author’s institution. We create the questionnaire through the Credamo platform and recruit 20 volunteers to participate in the survey, which is comparable to similar studies, e.g., AntiFake (24 participants) and E2E-VGuard (36 participants). All volunteers are over 18 years old and are English-educated. Their average response time is 333.000 seconds, providing reliable results.

Questionnaire Setup. We randomly select four speakers (two male and two female) from the SeedTTS-EN dataset, with one utterance per speaker. Each utterance is processed by three baselines, i.e., E2E-VGuard and AudioShield with two perturbation budgets $\tau = 0.1$ and $\tau = 0.3$, as well as our T-SemAttack, resulting in 16 attacked audio samples and 4 original samples. For each attacked sample, participants are asked to compare it with the corresponding original audio and provide ratings from three perspectives: (1) **Audio Quality**, which measures the noise level and overall perceptual quality of the attacked audio; (2) **Content Consistency**, which evaluates whether the linguistic content of the attacked audio remains aligned with the original; and (3) **Timbre Similarity**, which assesses whether the speaker identity in the attacked audio is preserved relative to the original.

All three aspects are scored on a 5-point Mean Opinion Score (MOS) scale, where higher scores indicate better perceptual quality. To improve the confidence of the subjective evaluation and reduce potential bias, we report the MOS values together with their 95% confidence intervals [29].

Table 9: Subjective evaluation from audio quality, content intelligibility, and speaker similarity aspects with 95% confidence intervals.

Method	MOS-Q($\uparrow$)	MOS-C($\uparrow$)	MOS-S($\uparrow$)
E2E-VGuard	2.67 $\pm$ 0.21	4.29 $\pm$ 0.17	3.98 $\pm$ 0.20
AudioShield ($\tau = 0.1$)	2.63 $\pm$ 0.30	3.32 $\pm$ 0.31	1.66 $\pm$ 0.33
AudioShield ($\tau = 0.3$)	1.39 $\pm$ 0.25	2.34 $\pm$ 0.35	1.38 $\pm$ 0.31
T-SemAttack	3.01$\pm$0.28	4.32$\pm$0.19	3.87$\pm$0.25

Results and Analysis. The detailed MOS results and analysis are provided in Table 9. We evaluate three dimensions: overall quality (MOS-Q), content intelligibility (MOS-C), and speaker similarity (MOS-S). Across all three dimensions, our method, T-SemAttack, consistently achieves the highest or comparable MOS values among all methods. In particular, AudioShield with $\tau = 0.3$ introduces a

clearly perceivable perturbation that almost prevents listeners from recognizing the linguistic content of the original audio, as reflected by its low MOS-Q (1.39), MOS-C (2.34), and MOS-S (1.38). Even with a smaller budget of $\tau = 0.1$, AudioShield still alters the timbre of certain audio samples, yielding a particularly low MOS-S of 1.66 and a degraded MOS-C of 3.32. We attribute this to the fact that AudioShield is trained on LibriSpeech and has fitted a substantial amount of female voice characteristics, and that modifying the latent representation cannot guarantee the fidelity of the reconstructed waveform. E2E-VGuard better preserves content intelligibility with a MOS-C of 4.29 and a MOS-S of 3.98 for speaker similarity, but still introduces audible artifacts that lower its overall quality with a MOS-Q of 2.67. In contrast, T-SemAttack adopts a linear additive strategy combined with the psychoacoustic masking effect, which minimizes its impact on the original audio content. It achieves the highest MOS-Q (3.01), which surpasses the 3.0 threshold generally regarded as acceptable perceptual distortion, and maintains strong MOS-C (4.32) and MOS-S (3.87). These subjective experiments show that T-SemAttack does not compromise the linguistic content or speaker timbre, thereby preserving the normal usability of the audio.

T-SemAttack Human study

This questionnaire is designed to assess the perceived quality of a series of audio samples. You will hear several groups of audio samples, each containing one original reference audio and several processed audio clips. Please listen carefully to each audio clip in a quiet environment, using headphones or high-quality playback equipment. Rate each clip based on the following three dimensions (on a scale of 0 to 5, with one decimal place allowed):

- Audio Quality: Whether the audio sounds clear and natural overall, and whether there are issues such as noise, distortion, or other problems affecting the listening experience. (0 = Very poor, 5 = Excellent)
- Content Match: Whether the spoken content of the audio matches the given text, and whether every word can be heard clearly. (0 = Completely unintelligible / No match at all, 5 = Completely clear / Perfect match)
- Timbre Consistency: (Compared with the first original audio in each group) Whether the timbre and voice characteristics of the speaker are consistent with the original audio. (0 = Completely different, 5 = Completely identical)

We will not collect your personal information. By completing this questionnaire, you consent to the use of your responses for academic research purposes.

21 original audio

original text: In other words, these three men took down the lectures which Luther addressed to his students in the course of galatians, and Roerer prepared the manuscript for the printer.

▶ 0:00 / 0:08 🔊 ⋮

22 Play the following audio and compare it with the original reference audio and the text, then rate it based on the following three dimensions: audio quality, content match, and timbre consistency.

▶ 0:00 / 0:08 🔊 ⋮

Q6 Rate the audio based on the following three dimensions: audio quality, content match, and timbre consistency (0 = Very poor, 5 = Excellent).

	0	1	2	3	4	5
Whether the audio sounds clear and natural overall.	●					0
Whether the spoken content of the audio matches the given text.	●					0
Whether the timbre and voice characteristics of the speaker are consistent with the original audio.	●					0

Figure 5: The screenshot of the questionnaire for subjective experiments.

The screenshot and full text of the questionnaire are shown in Fig. 5.

F Additional Details

F.1 Attack Details

Table 10 summarizes the main hyperparameters used throughout T-SemAttack. “lr” refers to the initial learning rate of the SGD optimizer, used to control the update step size of the adversarial noise during backpropagation. “lr_min” refers to the minimal learning rate. “max_steps” denotes the total optimization steps for the T-SemAttack. ϵ represents the perturbation amplitude.

Table 10: Hyperparameters of T-SemAttack.

Hyperparameter	Value
lr	1e-3
lr_min	1e-6
max_steps	500
ϵ	8/255

F.2 Adversarial Defense Details

The following are the details of adversarial defense and audio transformations for Table 4 we use against semantic attacks:

- **Resample.** Following standard adversarial defense practice [33], we randomly downsample the input waveform to one of 8000, 10000, 12000 Hz and then upsample it back to the original 24 kHz. This destroys high-frequency components where adversarial perturbations are often concentrated.
- **Quantize and Dequantize (Q-D).** The waveform is linearly mapped from the $[-1, 1]$ range to a b-bit integer range and then mapped back to its original scale. This bit-depth reduction acts as a coarse projection that disrupts small adversarial perturbations in the waveform amplitude.
- **Mel-Inverse Reconstruction (Mel).** The waveform is first transformed into a mel-spectrogram and then reconstructed back to the time domain using a Griffin-Lim-based inversion. This process is lossy and discards the original phase information, acting as a strong non-linear filter that removes perturbations not aligned with the mel-scale spectral structure. The configuration uses an FFT size of 2048, a hop length of 512, 128 mel bands, and a sampling rate of 24 kHz.
- **AudioPure** [34]. AudioPure is a defense method against adversarial audio attacks. It uses a diffusion model to purify perturbed audio by adding noise and then denoising it through the reverse diffusion process, reducing adversarial perturbations and making the audio closer to clean speech, thereby improving the robustness of speech recognition or audio classification systems under attack.
- **Change speed (Speed).** We apply time stretching with a randomly selected rate from the set 0.8, 0.9, 1.0, 1.1, 1.2. This temporally realigns acoustic events without altering pitch.
- **MP3 Compression (MP3).** We apply MP3 compression with bitrates uniformly sampled between 8 and 64 kbps while preserving the original waveform length. The compressed signal is either truncated or zero-padded as needed to maintain alignment.
- **Pitch Shift (PS).** We shift the pitch by a random number of semitones uniformly sampled from -4 to +4. This modifies the pitch contour while preserving the duration of the audio.
- **Band-Pass Filter (BPF).** We apply a band-pass filter with a center frequency uniformly sampled between 100 and 6000 Hz. This restricts the spectral content of the input, retaining only a specific frequency band.

The adversarial defense methods follow WaveGuard [33]. Audio transformation techniques are implemented using the audiomentations library

F.3 ROSETok and Training Details

Details of the ROSETok Architecture. The ROSETok framework comprises several core components: an encoder compresses the input audio waveform into a 1280-dimensional latent representation, which is then aligned with semantic embeddings from a Whisper Teacher model. A 9-codebook Residual Vector Quantizer subsequently discretizes this latent representation into compact codes, and a decoder reconstructs the waveform from these codes.

The DAC encoder and decoder are constructed from symmetrically arranged downsampling and upsampling modules. The encoder contains four EncoderBlocks that progressively downsample the signal with striding factors of 2, 4, 5, and 8, while expanding the channel dimension from 64 to 1024 and finally projecting to a 1280-dimensional latent space. Each EncoderBlock sequentially connects three residual units with dilation rates of 1, 3, and 9, followed by a Snake activation function. The decoder mirrors this architecture exactly, starting with 1536 channels and employing four DecoderBlocks with upsampling factors of 8, 5, 4, and 2 to restore the monaural audio waveform.

The quantizer adopts residual vector quantization with nine codebook layers, where each layer maintains 1024 entries of 8-dimensional vectors. The quantization dropout rate is set to zero to ensure training stability. Through successive residual refinement across codebook layers, a compact 8-dimensional discrete representation approximates the original 1280-dimensional continuous latent vector, achieving a compression ratio of approximately 17.8.

During the training stage, two additional components are introduced, namely the semantic encoder and the discriminator. During the training process, the semantic encoder employs a frozen Whisper large-v3 encoder, which first extracts 80-dimensional Mel spectrogram features via short-time Fourier transform and then produces 1280-dimensional semantic embeddings through its Transformer backbone. This embedding dimensionality aligns precisely with the output dimension of the DAC encoder. Additionally, the discriminator is introduced to enhance the perceptual quality of the reconstructed audio by comparing it against the original audio, where adversarial loss and feature matching loss are computed to guide the training.

Details of Training Objectives. The details of each loss during training are as follows:

$\mathcal{L}_{\text{distill}}$ is the semantic distillation loss, which forces the audio encoder to learn semantically rich representations by maximizing the cosine similarity between the latent representations output by the encoder and the semantic embeddings extracted by the frozen Whisper teacher model [18]. The semantic distillation loss $\mathcal{L}_{\text{distill}}$ is computed as $\mathcal{L}_{\text{distill}} = 1 - \text{CosSim}(z, z_{\text{sem}})$, where z denotes the encoded representation and z_{sem} is the representation from the semantic encoder.

$\mathcal{L}_{\text{recon}}$ is the reconstruction loss, which ensures the fidelity of the codec process by minimizing the difference between the audio reconstructed by the decoder and the original audio.

$\mathcal{L}_{\text{cb}}$ is the codebook loss, which optimizes the codebook vectors of the residual vector quantizer to better fit the feature distribution output by the encoder.

$\mathcal{L}_{\text{commit}}$ is the commitment loss, which constrains the encoder output to stay close to the nearest codebook vector, preventing large fluctuations between different codebooks.

$\mathcal{L}_{\text{adv}}$ is the adversarial loss, which improves the perceptual quality of the generated audio by training a discriminator to distinguish between real and reconstructed audio.

$\mathcal{L}_{\text{fm}}$ is the feature matching loss, which stabilizes the adversarial training process and further enhances the quality of the reconstructed audio by minimizing the difference between features extracted from the original and reconstructed audio at intermediate layers of the discriminator.

We adopt $\mathcal{L}_{\text{recon}}$, $\mathcal{L}_{\text{cb}}$, and $\mathcal{L}_{\text{commit}}$ from [23], and $\mathcal{L}_{\text{adv}}$ and $\mathcal{L}_{\text{fm}}$ from [24].

Details of Training Datasets. We train the ROSETok model on 96.7k hours of open-source English and Chinese data. All open-source datasets used in this work are listed in Table 11. LibriSpeech is used only for ablation and is not counted in the 96.7k-hour ROSETok pretraining corpus.

Table 11: Detailed information of training datasets.

Dataset	Subset	Duration (# h)	Task	Language	Usage
Librispeech [37]	train & val	960	ASR	English	Ablation
Emilia [30]	EN	46,800	ASR	English	ROSETok
Emilia [30]	ZH	49,900	ASR	Chinese	ROSETok

Details of Hyperparameters. Table 12 summarizes the main hyperparameters used throughout ROSETok training. “lr” refers to the initial learning rate of the AdamW optimizer, controlling the step size of parameter updates during training. “lr_scheduler” refers to the learning rate scheduling strategy, where CosineLR is used to smoothly anneal the learning rate following a cosine curve. “batch_size” refers to the global number of audio samples per training step. “noise_level” refers to the noise amplitude during noisy training. “warmup_steps” refers to the initial 1000 steps during which the learning rate is linearly warmed up to stabilize early-stage training. “total_steps” refers to the total number of optimization steps 200,000 for training the robust audio tokenizer.

Details of the XARES Benchmark. XARES [38] evaluates audio encoders through downstream probing tasks. We freeze the encoder, extract frame-level speech representations, and feed them into lightweight task-specific MLP heads for classification. Following the XARES protocol, raw

Table 12: Training Hyperparameters of the ROSETok.

Hyperparameter	Value
optimizer	AdamW
lr	1.0×10^{-4}
lr_scheduler	CosineLR
min_lr	1.0×10^{-5}
batch_size	128
noise_level	$4/255 \sim 24/255$
warmup_steps	1000
total_steps	200,000

task metrics are normalized to $[0, 1]$, where higher scores indicate stronger semantic capability. For readability, we report the normalized scores in percentage form.

F.4 Baseline and Model Details

The baseline attack methods considered in our study are as follows: (1) **Gaussian Noise**. As the most fundamental comparison baseline, this method introduces perturbations by adding random Gaussian noise to the original audio signal. To ensure that the signal-to-noise ratio of the processed audio is comparable to that of our proposed method, facilitating a fair comparison, we set the noise standard deviation to 0.01. (2) **E2E-VGuard** [7] is an end-to-end adversarial defense scheme designed for speech models. In our experiments, we strictly follow the default parameter configurations from the original paper to ensure the reliability of reproduction results and the objectivity of the comparison. (3) **AudioShield** [8] attacks audio encoders by applying perturbations in the latent feature space, where a threshold parameter τ controls the trade-off between perturbation strength and perception. To thoroughly examine the performance of this method under different perturbation intensities, we set up two sets of comparative experiments with $\tau = 0.3$ and $\tau = 0.1$, corresponding to two typical scenarios of strong perturbation and weak perturbation, respectively.

To comprehensively evaluate the effectiveness of our semantic attack, we test the adversarial audio against a diverse set of Automatic Speech Recognition (ASR) models spanning both open-source and closed-source systems. The open-source models we use are as follows:

(1) **HuBERT-Large** [17] is a self-supervised speech representation model fine-tuned on 960 hours of LibriSpeech data, known for its strong performance on English transcription tasks through masked prediction of hidden units. (2) **Whisper-Large** [18] is OpenAI’s multilingual, multitask ASR model trained on 680k+ hours of weakly supervised audio data, offering robust performance across languages and acoustic conditions. HuBERT-Large and Whisper-Large are evaluated as ASR systems in Table 2, while their encoders are also used as surrogate branches during attack optimization. (3) **Qwen3-ASR-0.6B** [6] is Alibaba’s lightweight ASR model built upon the Qwen3 foundation, providing efficient and accurate multilingual speech recognition with a compact 0.6B parameter footprint. (4) **VibeVoice-ASR** [26] is a recent ASR system designed for natural conversational speech, emphasizing robustness to spontaneous and expressive vocal styles. (5) **Fun-ASR-Nano** [28] is a compact variant of the FunASR family developed by Alibaba DAMO Academy, optimized for low-latency inference while maintaining competitive accuracy on Mandarin and multilingual benchmarks. (6) **FireRedASR** [27] is an attention-based encoder-decoder ASR model from Xiaohongshu, achieving SOTA results on Mandarin benchmarks with strong generalization to diverse acoustic scenarios.

For closed-source ASR systems, we evaluate against three widely-used commercial APIs: iFlytek, Doubao, and GPT-4o-mini-transcribe, which allows us to assess the transferability of our attack to black-box production systems. (1) **iFlytek** is a leading Chinese commercial ASR service renowned for its high accuracy on Mandarin and various dialects in industrial deployments. (2) **Doubao-ASR** is ByteDance’s commercial speech recognition API, offering robust transcription capabilities across multiple languages and noisy real-world conditions. (3) **GPT-4o-mini-transcribe** is OpenAI’s lightweight transcription endpoint derived from the GPT-4o family, providing high-quality multilingual transcription through a unified multimodal architecture.

In addition to dedicated ASR systems, we further evaluate our attack against LALMs, which integrate audio understanding with large language model capabilities. (1) **Qwen2-Audio** [1] is Alibaba’s instruction-tuned audio-language model that supports both voice chat and audio analysis, capable of processing diverse audio inputs including speech, sounds, and music. (2) **Qwen3-Omni** [2] is a flagship omni-modal model with a Mixture-of-Experts architecture (30B total / 3B active parameters), unifying text, image, audio, and video understanding with strong instruction-following capabilities. (3) **Audio-Flamingo-3** [5] is NVIDIA’s advanced audio-language model that excels at understanding complex audio content, including speech, music, and ambient sounds, with strong reasoning and dialogue capabilities. The closed-source category includes **Gemini-3-Flash-Preview**, Google’s latest multimodal model from the Gemini family, offering fast inference and strong cross-modal understanding across audio, vision, and text.

To ensure these models perform pure transcription rather than higher-level reasoning or summarization, we adopt a unified system prompt of “*Transcribe the audio exactly as recorded. Output the transcription only. Do not add explanations, summaries, or comments.*” This setup enables a fair comparison between LALMs and conventional ASR models under the same transcription objective.

To evaluate the effectiveness of our noise-robust tokenizer, we compare it against a diverse set of representative baseline tokenizers. (1) **SNAC** [35] is a multi-scale neural audio codec that employs hierarchical RVQ at different temporal resolutions to efficiently compress audio signals. (2) **XY-Tokenizer** [36] is a speech tokenizer designed to jointly model semantic and acoustic information, enabling unified representation for speech understanding and generation. (3) **XCodec-2** [24] integrates semantic features from self-supervised models with acoustic reconstruction objectives, producing discrete tokens that balance linguistic content and audio fidelity. (4) **DAC** [23] is a high-fidelity general-purpose neural audio codec that leverages improved RVQ techniques and adversarial training to achieve strong reconstruction quality across diverse audio domains. (5) **MOSS-Audio-Tokenizer** [22] is a neural audio codec that compresses raw audio into discrete tokens, enabling efficient audio generation and understanding within the MOSS multimodal framework. (6) **StableToken** [15] is a tokenizer that focuses on producing stable and consistent discrete representations across varying input conditions, aiming to mitigate token instability in downstream tasks. Together, these baselines cover a broad spectrum of design philosophies, ranging from purely acoustic codecs to semantically-enriched and stability-oriented tokenizers, providing a comprehensive baseline system for assessing noise robustness.

F.5 Compute Reporting and Efficiency Details

Details of Attacks’ Efficiency Evaluation. In Table 2, the RTF is computed as the ratio of the per-sample perturbation generation time to the total duration of the audio in Eq. (6).

$$\text{RTF}_{\text{Attacks}} = \frac{\sum_{i=1}^N T_{\text{attack}}^{(i)}}{\sum_{i=1}^N T_{\text{audio}}^{(i)}}, \quad (6)$$

where $T_{\text{attack}}^{(i)}$ is the time taken by each attack method to process the i -th audio sample, $T_{\text{audio}}^{(i)}$ is the duration of the i -th audio sample, and N is the total number of test samples.

AudioShield constitutes a special case because it pre-trains a universal adversarial perturbation (UAP) on a training set, then directly applies it to all test inputs without per-audio optimization. Since the training stage is essential for obtaining a usable perturbation, we amortize its cost across test samples to ensure a fair comparison in Eq. (7).

$$\text{RTF}_{\text{AudioShield}} = \frac{T_{\text{train}} + \sum_{i=1}^N T_{\text{apply}}^{(i)}}{\sum_{i=1}^N T_{\text{audio}}^{(i)}}, \quad (7)$$

where T_{train} is the total perturbation training time, $T_{\text{apply}}^{(i)}$ is the time to apply the pre-trained perturbation to the i -th audio sample. This amortizes the training overhead across test samples, enabling a fair comparison with other baselines under a unified framework. When N is sufficiently large, the amortized T_{train} term becomes negligible, and the RTF of AudioShield gradually approaches its theoretical lower bound. The RTF measurements of attacks are conducted on a device equipped with an NVIDIA 4090 GPU and Intel(R) Xeon(R) Platinum 8488C CPU, 512GB RAM.

843 **Details of Tokenizers’ Efficiency Evaluation.** In Table 5, the Real-Time Factor (RTF) is computed
 844 as the ratio of the total audio reconstruction time to the total duration of the audio in Eq. (8).

$$\text{RTF}_{\text{Tokenizers}} = \frac{\sum_{i=1}^N T_{\text{reconstruct}}^{(i)}}{\sum_{i=1}^N T_{\text{audio}}^{(i)}}, \quad (8)$$

845 The reconstruction time $T_{\text{reconstruct}}^{(i)}$ measures only the tokenizer’s core reconstruction process, namely
 846 waveform encoding and decoding. It excludes the time required to load audio files from disk or save
 847 reconstructed audio back to disk. This ensures that the reported RTF reflects the robust tokenizer’s
 848 intrinsic computational efficiency, rather than I/O overhead unrelated to model performance. The
 849 RTF measurements of tokenizers are conducted on a device equipped with an NVIDIA 4090 GPU,
 850 Intel(R) Xeon(R) Platinum 8488C CPU, and 512 GB RAM.

851 **Compute Reporting.** We conduct our experiments using GPU-equipped computing resources.
 852 Specifically, we run T-SemAttack and the evaluation on a single NVIDIA RTX 4090 GPU. For
 853 training ROSETok, we use 8 NVIDIA H100 GPUs for approximately 20 hours.

854 G Visualization and Analyses of the T-SemAttack

855 G.1 Perturbation Amplification

856 To analyze the mechanism of T-SemAttack in the speech tokenizer, we compute the representation
 857 differences between clean audio x and adversarial audio x_{adv} at different internal stages of the
 858 tokenizer. We use the S3 Tokenizer as an example, and its audio processing pipeline is: Waveform $\rightarrow$
 859 Mel-spectrogram $\rightarrow$ Encoder $\rightarrow$ VQ Tokens.

860 For continuous representation stages, we use relative L_2 distortion, referred to as Relative Distortion
 861 (RD), to measure the perturbation magnitude:

$$\text{RD}_s = \frac{\|z_s(x_{\text{adv}}) - z_s(x)\|_2}{\|z_s(x)\|_2},$$

862 where $z_s(\cdot) \in \mathbb{R}^{T_s \times d_s}$ denotes the representation at stage s , T_s is the sequence length of the
 863 representation at that stage, and d_s is the feature dimension. Note that T_s is stage-dependent and
 864 may differ from the waveform length T . We also use cosine distance (CD) to measure changes in the
 865 representation direction:

$$\text{CD}_s = 1 - \frac{1}{T_s} \sum_{t=1}^{T_s} \cos(z_{s,t}(x), z_{s,t}(x_{\text{adv}})).$$

866 The amplification factor at each stage is defined as the ratio between the average perturbation at that
 867 stage and the average perturbation at the waveform stage:

$$\text{Amp}_{\text{RD}}(s) = \frac{\frac{1}{N} \sum_{i=1}^N \text{RD}_s^{(i)}}{\frac{1}{N} \sum_{i=1}^N \text{RD}_{\text{wav}}^{(i)}}.$$

868 For the final discrete VQ tokens, we compute the token error rate (TER):

$$\text{TER} = 1 - \frac{\sum_{n=1}^N \mathbf{1}[c_n(x) = c_n(x_{\text{adv}})]}{N},$$

869 where $c_n(\cdot)$ denotes the n -th discrete token and N denotes the number of tokens.

870 Table 13 reports the perturbation amplification results on the S3 Tokenizer. The average RD at
 871 the waveform stage is only 0.154, but it rapidly increases to 0.771 at the Mel-spectrogram stage,
 872 corresponding to a $5.01\times$ amplification. After the encoder, the average RD further increases to 1.520,
 873 which is $9.87\times$ larger than that at the waveform stage. Meanwhile, the average CD of the encoder
 874 output reaches 1.248, indicating a severe directional shift in the high-level semantic representations
 875 of the adversarial samples compared with the clean samples.

876 Finally, at the VQ token level, T-SemAttack achieves an average TER of 99.7%. This shows that
 877 perturbation drift in the continuous representation space is ultimately quantized into an almost entirely
 878 different discrete token sequence. Therefore, T-SemAttack does not merely introduce local waveform
 879 noise. Instead, it creates a progressive amplification chain inside the S3 Tokenizer, from the acoustic
 880 frontend to the semantic encoder, ultimately causing tokenizer-level semantic collapse.

Table 13: Perturbation amplification results in the S3 Tokenizer.

Stage	RD Mean	RD Std	RD Median	CD Mean	CD Median	Amp(RD)
Waveform	0.154	0.023	0.153	0.322	0.326	1.00×
Mel Spectrogram	0.771	0.070	0.772	0.357	0.347	5.01×
Encoder Output	1.520	0.055	1.530	1.248	1.250	9.87×

G.2 LALM Attention

To further analyze the impact of T-SemAttack on the cross-modal modeling process of LALMs, we take Qwen2-Audio as an example and conduct a layer-wise analysis of Transformer attention under clean and adversarial audio inputs. The analysis consists of a multi-sample statistical analysis on LibriTTS, corresponding to the audio attention ratio in Fig. 2b, and a single-sample attention-map visualization, which intuitively shows changes in the attention distribution before and after the attack.

Multi-Sample Attention Ratio. For each clean/adversarial audio pair, we concatenate the audio with its corresponding transcription as the same model input:

$$\mathbf{x} = \langle \text{audio_bos} \rangle \langle \text{AUDIO} \rangle \langle \text{audio_eos} \rangle$$

Detect the language and recognize the speech: $\langle \text{len} \rangle + y$,

where y denotes the ground-truth transcription of the audio. During the forward pass, we enable `output_attentions=True` to obtain the attention matrix for each layer and each attention head. For the l -th layer, we first average over all heads as $A_l = \frac{1}{H} \sum_{h=1}^H A_{l,h}$, where H is the number of attention heads, and $A_l \in \mathbb{R}^{T \times T}$ denotes the head-averaged attention matrix.

We then locate the audio-token interval I_a using $\langle \text{audio_bos} \rangle$ and $\langle \text{audio_eos} \rangle$ in the input sequence, locate the target-text interval I_t according to the token length of the target transcription, and define the tokens between the end of the audio segment and the start of the target text as the prompt interval I_p . For the i -th sample and the l -th layer, we compute the average attention from target-text query tokens to audio keys as $s_{l,\text{audio}}^{(i)} = \frac{1}{|I_t||I_a|} \sum_{q \in I_t} \sum_{k \in I_a} A_l^{(i)}[q, k]$.

Similarly, we compute the average attention from target tokens to prompt tokens and target tokens as $s_{l,\text{prompt}}^{(i)}$ and $s_{l,\text{target}}^{(i)}$. Fig. 2b in the main paper shows the normalized share of attention assigned to the audio input at each layer. This value is computed by comparing the attention given to audio against the total attention given to audio, prompt, and target tokens, with a small constant added for numerical stability. This metric measures the relative proportion of attention assigned to audio tokens among the audio, prompt, and target contexts during target-text modeling. We compute $r_l^{(i)}$ for clean and adversarial samples separately, and report the sample mean and standard deviation at each layer as $\bar{r}_l = \frac{1}{N} \sum_{i=1}^N r_l^{(i)}$. The curves in Fig. 2b correspond to $\bar{r}_l$.

The results show that T-SemAttack substantially reduces the attention ratio from target-text tokens to audio tokens in Qwen2-Audio’s middle and upper layers. Around layers 9–28, the audio attention ratio under adversarial audio consistently decreases relative to clean audio. These observations suggest that, after the attack, the model relies less on audio context when recognizing or generating the target text and instead depends more on the prompt or target-side self-context. In other words, T-SemAttack not only disrupts semantic representations produced by the speech tokenizer but also further interferes with the LALM’s internal cross-modal alignment process.

Single-Sample Attention Visualization. To further illustrate attention disruption, we select one clean-adversarial pair and visualize the attention maps of several representative layers. For each layer, we average over all attention heads and then plot the resulting two-dimensional attention matrix. In the figure, the x-axis denotes key positions, and the y-axis denotes query positions. Since the LLM component of Qwen2-Audio uses causal attention, the future-token region in the upper-right corner is masked, appearing as a black triangular area.

Fig. 6 presents a comparison between clean and adversarial attention maps. Brighter colors correspond to larger attention weights from the query token to the key token. To display both strong and weak attention regions, we use log-scale normalization in the visualization.

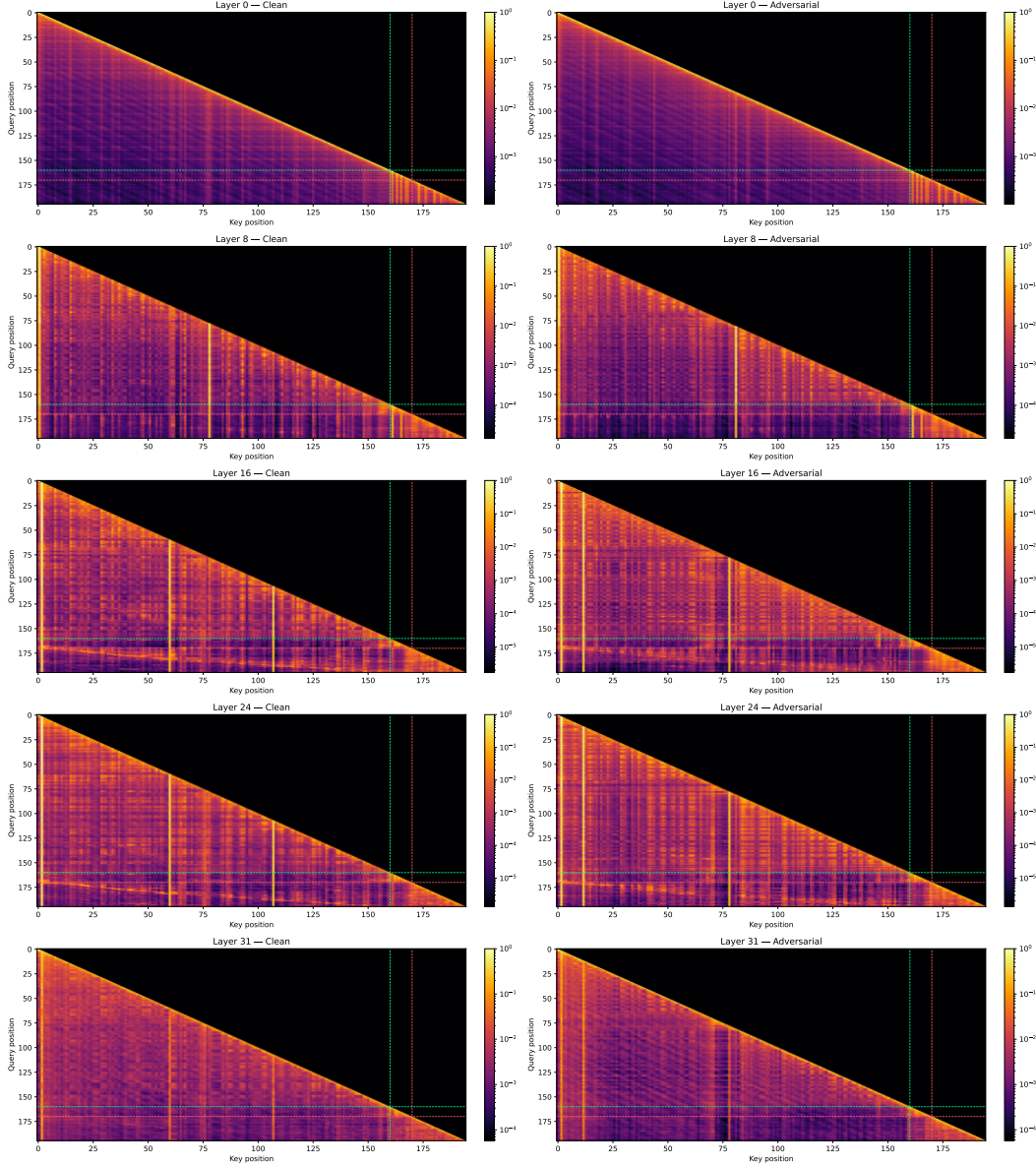


Figure 6: Single-sample attention maps of Qwen2-Audio under clean and adversarial inputs. Each row corresponds to one Transformer layer, while the left and right columns show clean and adversarial maps, respectively. Green dashed lines mark the audio-token boundaries, and red dashed lines mark the beginning of the target-text region.

922 To better observe the attention changes caused by the attack, we further plot the differential attention
 923 map as $\Delta A_l = A_l^{\text{adv}} - A_l^{\text{clean}}$. Fig. 7 shows the differential maps across different layers. Red
 924 indicates increased attention under the adversarial condition, while blue indicates decreased attention.

925 The differential maps show that attention changes are more pronounced in the middle and upper
 926 layers, and are concentrated near the audio boundaries, the target-text start position, and several
 927 key positions strongly relied on by the model. This is consistent with the multi-sample statistical
 928 results in Fig. 2b: T-SemAttack systematically reduces the relative attention from target tokens to
 929 audio tokens, causing the LALM to rely more on target-side context rather than the original audio
 930 information during text modeling.

931 Therefore, the multi-sample statistics and single-sample visualization provide complementary visu-
 932 alizations. Fig. 2b shows at the aggregate level that T-SemAttack reduces the attention allocation

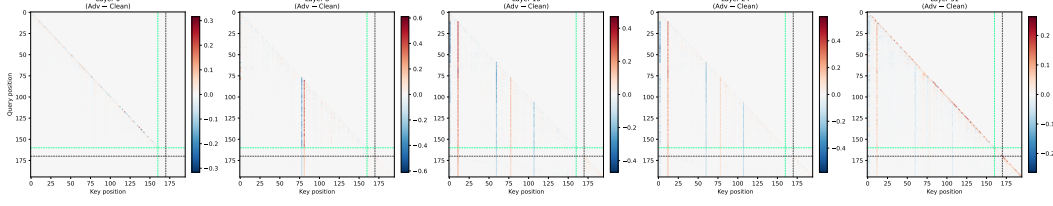


Figure 7: Single-sample attention difference maps between adversarial and clean inputs. The visualized value is $\Delta A_l = A_l^{\text{adv}} - A_l^{\text{clean}}$. Red regions indicate increased attention under adversarial input, while blue regions indicate decreased attention.

of the LALM to audio tokens, while Fig. 6 and Fig. 7 illustrate how this change appears in specific samples, layers, and token regions. These observations suggest that once the semantic representations of the speech tokenizer are disrupted by our semantic attack, the cross-modal alignment process of the LALM is also affected, making it difficult for the model to recover the correct semantics from the audio context and ultimately leading to understanding degradation.

H Transcription Examples

We provide some attack examples in Table 14.

Table 14: Transcriptions on different ASR models

Method	WER (%)	Transcriptions
Whisper Large		
Original	–	roaming endlessly around the park she wants to go home
Clean	0.000	roaming endlessly around the park she wants to go home
Gaussian	10.000	blooming endlessly around the park she wants to go home
E2E-VGuard	10.000	roaming endlessly around the park he wants to go home
AudioShield-0.1	90.000	roman did not listen to the wrong that hour she wanted to go home
AudioShield-0.3	100.000	our little dosmoud around for there ar around for now on
T-SemAttack	250.000	and olympics and are on the path to a free world home have fun watching people ? have fun watching people ? have fun watching people ?
VibeVoice-ASR		
Original	0.000	i am never more aware of a room is acoustics than when i am trying to enjoy a snack i have no intention of sharing
Gaussian	12.000	i am never more aware of a room is acoustics than when i am trying to listen back i have no intention of sharing
E2E-VGuard	64.000	i am never going to really believe the postscript and i am calling you just now i have no intention to
AudioShield-0.1	28.000	i am never more aware of a woman is acoustics than when i am trying to enjoy a snack i have not looked into the tree already
AudioShield-0.3	64.000	i am never aware of the most coast things that i am planning to listen to i have nothing to share with you
T-SemAttack	116.000	i never know where to leave i do not think it is going to be fun it is going to be good i hope you do not think it is too late
Qwen3-ASR		
Original	0.000	i am never more aware of a room is acoustics than when i am trying to enjoy a snack i have no intention of sharing
clean	4.000	i am never more aware of a room is coziest than when i am trying to enjoy a snack i have no intention of sharing
Gaussian	4.000	i am never more aware of a room is coasticks than when i am trying to enjoy a snack i have no intention of sharing
E2E-VGuard	44.000	i am never more aware of a wind is approach than when one gives me a snack i have made them so care
AudioShield-0.1	28.000	i am never more aware of romanticostics than when trying to eat a snack i have no intention of sharing
AudioShield-0.3	76.000	i am having a weird weird cold did you understand why i am suddenly getting worse i have i have a little shivering
T-SemAttack	88.000	i remember we are going to coast beach in lantau the green map i have used on the computer

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